WORKED SOLUTION

STEP III 2018, Question 4

Pure • Tangents to a Hyperbola

STEP III 2018, Question 4

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s3-q4

The key idea

The tangent is quickest from the standard result that the tangent to $x^2/a^2 - y^2/b^2 = 1$ at (x_1, y_1) is $xx_1/a^2 - yy_1/b^2 = 1$. After that, everything is two straight lines meeting, so it is a matter of solving pairs of linear equations and tidying the trigonometry. In part (ii) the perpendicularity condition is a single relation between $\sin\theta$ and $\sin\phi$ involving b, and it is exactly what lets us remove b from the expressions and then bring it back as $a^2 - b^2$. Sum-to-product formulae make the final algebra short.

Solution

The tangent at P

Differentiating $x^2/a^2 - y^2/b^2 = 1$ implicitly gives $\dfrac{2x}{a^2} - \dfrac{2y}{b^2}\dfrac{\mathrm{d}y}{\mathrm{d}x} = 0$, so the gradient at (x_1, y_1) is $\dfrac{b^2x_1}{a^2y_1}$ and the tangent is

$$y - y_1 = \frac{b^2x_1}{a^2y_1}(x - x_1) \iff \frac{xx_1}{a^2} - \frac{yy_1}{b^2} = \frac{x_1^2}{a^2} - \frac{y_1^2}{b^2} = 1,$$

using the fact that (x_1, y_1) is on the curve. (This form also covers the vertices, where $y_1 = 0$ and the tangent is vertical.) With $x_1 = a\sec\theta$, $y_1 = b\tan\theta$:

$$\frac{x\sec\theta}{a} - \frac{y\tan\theta}{b} = 1.$$

Multiplying through by $ab\cos\theta$ (which is non-zero, since $\sec\theta$ exists) gives

$$bx - ay\sin\theta = ab\cos\theta,$$

as required.

Part (i)

The two given lines are the asymptotes $y = \pm\frac{b}{a}x$.

At S, $ay = bx$. Substituting $ay = bx$ into the tangent gives $bx(1-\sin\theta) = ab\cos\theta$, so

$$S = \left(\frac{a\cos\theta}{1-\sin\theta},\ \frac{b\cos\theta}{1-\sin\theta}\right).$$

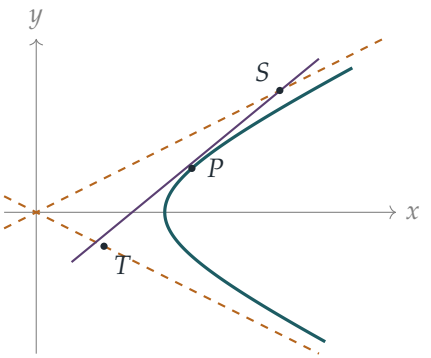
At T, $ay = -bx$, which gives $bx(1+\sin\theta) = ab\cos\theta$, so

$$T = \left(\frac{a\cos\theta}{1+\sin\theta},\ -\frac{b\cos\theta}{1+\sin\theta}\right).$$

Both denominators are non-zero because $\cos\theta \neq 0$ means $\sin\theta \neq \pm 1$. Adding, and using $(1-\sin\theta)(1+\sin\theta) = \cos^2\theta$:

$$x_S + x_T = a\cos\theta \cdot \frac{(1+\sin\theta)+(1-\sin\theta)}{\cos^2\theta} = \frac{2a}{\cos\theta} = 2a\sec\theta,$$
$$y_S + y_T = b\cos\theta \cdot \frac{(1+\sin\theta)-(1-\sin\theta)}{\cos^2\theta} = \frac{2b\sin\theta}{\cos\theta} = 2b\tan\theta.$$

So the mid-point of ST is $(a\sec\theta,\ b\tan\theta)$: **the mid-point of ST is P itself**. The tangent at any point is bisected at the point of contact by the two asymptotes.



Part (ii)

The perpendicularity condition. Written as $ay\sin\theta = bx - ab\cos\theta$, the tangent at P has gradient $\dfrac{b}{a\sin\theta}$ when $\sin\theta \neq 0$. (If $\sin\theta = 0$ the tangent is vertical, so the tangent at Q would have to be horizontal; but no tangent is horizontal, since $b/(a\sin\phi) \neq 0$. So $\sin\theta \neq 0$, and likewise $\sin\phi \neq 0$.) Perpendicular tangents need the product of the gradients to be -1:

$$\frac{b^2}{a^2\sin\theta\sin\phi} = -1 \qquad \Longleftrightarrow \qquad \sin\theta\sin\phi = -\frac{b^2}{a^2}. \tag{$*$}$$

In particular $\sin\theta$ and $\sin\phi$ have opposite signs, so $\sin\phi - \sin\theta \neq 0$. Note that $(*)$ needs $b^2/a^2 < 1$, which is where $a > b$ is used.

The intersection. The point (x, y) satisfies

$$bx - ay\sin\theta = ab\cos\theta, \qquad bx - ay\sin\phi = ab\cos\phi.$$

Subtracting the second from the first gives $ay(\sin\phi - \sin\theta) = ab(\cos\theta - \cos\phi)$. Multiplying the first by $\sin\phi$, the second by $\sin\theta$, and subtracting, gives $bx(\sin\phi - \sin\theta) = ab(\sin\phi\cos\theta - \cos\phi\sin\theta) = ab\sin(\phi - \theta)$. Hence

$$x = \frac{a\sin(\phi-\theta)}{\sin\phi - \sin\theta}, \qquad y = \frac{b(\cos\theta - \cos\phi)}{\sin\phi - \sin\theta}.$$

Ask yourself These still involve b. How can we write x^2 and y^2 using only a, θ and ϕ, and can the fractions be simplified?

The condition $(*)$ says $b^2 = -a^2\sin\theta\sin\phi$, so squaring removes b. To simplify, write $\alpha = \frac{1}{2}(\phi + \theta)$ and $\beta = \frac{1}{2}(\phi - \theta)$. The sum-to-product formulae give

$$\sin\phi - \sin\theta = 2\cos\alpha\sin\beta, \qquad \cos\theta - \cos\phi = 2\sin\alpha\sin\beta, \qquad \sin(\phi-\theta) = 2\sin\beta\cos\beta,$$

and $\cos\alpha\sin\beta \neq 0$ because $\sin\phi - \sin\theta \neq 0$. Cancelling $2\sin\beta$,

$$x = \frac{a\cos\beta}{\cos\alpha}, \qquad y = b\tan\alpha,$$

and therefore

$$x^2 = \frac{a^2\cos^2(\frac{1}{2}(\phi-\theta))}{\cos^2(\frac{1}{2}(\phi+\theta))},$$
$$y^2 = -a^2\sin\theta\sin\phi\,\tan^2(\tfrac{1}{2}(\phi+\theta)).$$

Equivalently, without half-angles,

$$x^2 = \frac{a^2\sin^2(\phi-\theta)}{(\sin\phi-\sin\theta)^2}, \qquad y^2 = -\frac{a^2\sin\theta\sin\phi\,(\cos\theta-\cos\phi)^2}{(\sin\phi-\sin\theta)^2}.$$

The circle. We also need $\sin\theta\sin\phi$ in terms of α and β. Since $\theta = \alpha - \beta$ and $\phi = \alpha + \beta$,

$$\begin{aligned}\sin\theta\sin\phi &= \sin^2\alpha\cos^2\beta - \cos^2\alpha\sin^2\beta \\ &= (1-\cos^2\alpha)\cos^2\beta - \cos^2\alpha(1-\cos^2\beta) = \cos^2\beta - \cos^2\alpha.\end{aligned}$$

Write $k = \sin\theta\sin\phi = \cos^2\beta - \cos^2\alpha$. Then, with $\tan^2\alpha = (1-\cos^2\alpha)/\cos^2\alpha$,

$$\begin{aligned}x^2 + y^2 &= \frac{a^2}{\cos^2\alpha}\Big[\cos^2\beta - k\,(1-\cos^2\alpha)\Big] \\ &= \frac{a^2}{\cos^2\alpha}\Big[(\cos^2\beta - k) + k\cos^2\alpha\Big] \\ &= \frac{a^2}{\cos^2\alpha}\Big[\cos^2\alpha + k\cos^2\alpha\Big] \qquad \text{since } \cos^2\beta - k = \cos^2\alpha \\ &= a^2(1+k) = a^2 + a^2\sin\theta\sin\phi = a^2 - b^2,\end{aligned}$$

using $(*)$ in the last step. So the point of intersection of any two perpendicular tangents lies on the circle $x^2 + y^2 = a^2 - b^2$, as required.

Checks

- Take $a = 2$, $b = 1$, $\theta = \frac{\pi}{6}$, $\phi = -\frac{\pi}{6}$. Then $\sin\theta\sin\phi = -\frac{1}{4} = -b^2/a^2$, so the tangents are perpendicular (gradients 1 and -1). Here $\alpha = 0$ and $\beta = -\frac{\pi}{6}$, so $x = 2\cos\frac{\pi}{6} = \sqrt{3}$ and $y = 0$, and $x^2 + y^2 = 3 = a^2 - b^2$. Directly: the tangents $x - y = \sqrt{3}$ and $x + y = \sqrt{3}$ do meet at $(\sqrt{3}, 0)$.
- In part (i), as $\theta \to 0$ the tangent becomes the vertical line $x = a$, which meets the asymptotes at $(a, \pm b)$, with mid-point $(a, 0) = P$.
- All the formulae above, including the mid-point in (i) and $x^2 + y^2 = a^2 - b^2$, were confirmed by computer for thousands of random choices of $a > b > 0$ and perpendicular pairs θ, ϕ, solving the two tangent equations directly.

Common slips

- Leaving b in the expressions for x^2 and y^2: the question asks for a, θ and ϕ only, so $(*)$ must be used to eliminate b^2.
- Taking the gradient of the tangent as $-b/(a\sin\theta)$ by mis-rearranging; check that it is positive at a point in the first quadrant.
- Dividing by $\sin\phi - \sin\theta$ without saying why it is non-zero.

Beyond STEP

The circle in part (ii) is the *director circle* of the hyperbola: the locus of points from which the two tangents are perpendicular. For an ellipse the director circle is $x^2 + y^2 = a^2 + b^2$, and for a hyperbola with $a < b$ there is no real director circle at all, which is why the question insists on $a > b$. The bisection property in part (i) is a first glimpse of projective and affine geometry of conics, where tangents, asymptotes and poles and polars are treated uniformly. For problems in exactly this spirit, try Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.